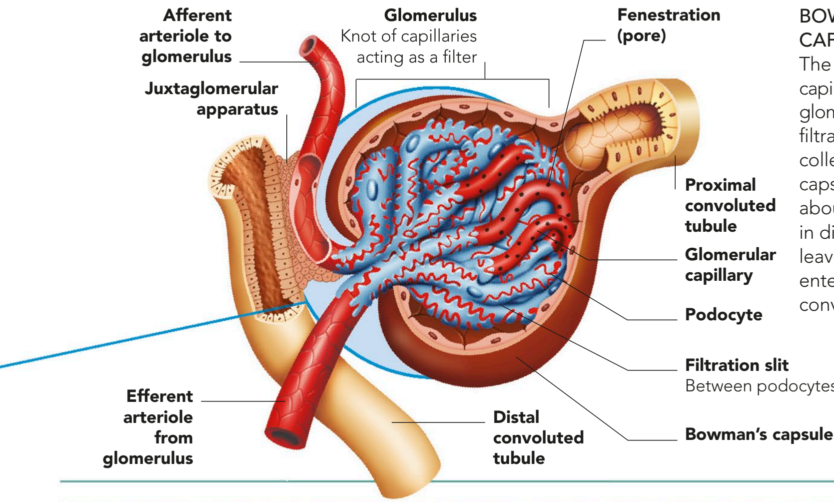

BOWMAN'S CAPSULE

The tangled system of capillaries forming the glomerulus oozes a filtrate fluid that is collected by Bowman's capsule. The capsule is about 0.2mm ($^{1}/_{125}$ in) in diameter. The filtrate leaves the capsule and enters the proximal convoluted tubule.

REGULATION OF URINE PRODUCTION

The amount, composition, and concentration of urine is determined principally by two hormones: ADH (antidiuretic hormone, or vasopressin) and aldosterone. ADH, released by the pituitary gland, acts on the kidneys to reduce urine volume and increase its concentration. Aldosterone, released by the adrenal glands, acts on the kidneys to reduce sodium and water in the urine and increase potassium.

HORMONAL CONTROL

Levels of the hormones ADH and aldosterone are altered so that the amounts of water, solutes, and wastes in urine are increased or decreased as needed to maintain a constant environment.